Simulation Concepts

Shen Cheng

ECP8506 Clinical Trial Simulation

10/30/2026

Outline

- Simulation Nomenclature
- Types of Simulations
- General Modeling Process
- General Simulation Process

Simulation Nomenclature

- Population parameter
  - Fixed effects (θ_n): parameters describing structural/covariate model
  - Random effects: parameters describing variability
    - Inter-individual/occasion variability (IIV/IOV, ω_n^2): Ω matrix ($\omega_1^2 \ldots \omega_n^2$)
    - Residual variability (RV, σ_n^2): Σ matrix ($\sigma_1^2 \ldots \sigma_n^2$)
- Individual IIV/IOV
  - Empirical Bayes estimates (EBEs): $\eta_{n,i} \sim N(0, \omega_n^2)$
- Individual RV: $\varepsilon_{n,ij} \sim N(0, \sigma_n^2)$
- Parameter uncertainty/precision
  - How confident/precise in estimating the population parameters
  - Relevant to both fixed effects & random effects
- Covariate: Patient characteristics (e.g., $\mathbf{weight_i}$, $\mathbf{height_i}$, $\mathbf{age_i}$, etc)
- Replicate: A single set of simulation incorporate all trial components

Annotations:
n: index of a parameter (e.g., the nth parameter).
i: index of an individual (e.g., the ith individual).
j: index of an observation (e.g., the observation collected at j^{th} time point).

Types of Simulations

- Deterministic vs. Stochastic
- Patient-level vs Population-level
- Drug Property vs Clinical Trial

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Deterministic vs. Stochastic Simulation

- Deterministic
  - Produce a typical response for a patient/population.
  - Do not need random number generator for Monte Carlo sampling.
- Stochastic (Monte Carlo)
  - Produce a distribution of response for a patient/population.
  - Require random number generator for Monte Carlo sampling in single or multiple levels:
    - Inter-individual / occasion variability
    - Parameter uncertainty

Patient-level vs Population-level

- Patient-level
  - Each simulation replicate correspond to a (typical) patient
- Population-level
  - Each simulation replicate correspond to a population (e.g., a trial patient population, a study arm, etc)

Drug Property vs Clinical Trial

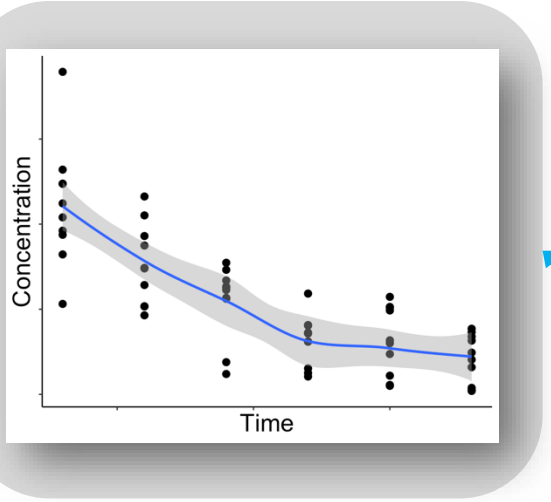
- Drug Property
  - To understand the property of a drug
  - Independent of study design
  - Example: what is the probability to achieve X response in a typical patient given Y dosing regimen of drug Z?
- Clinical Trial
  - To understand the performance of a trial design
  - Dependent on trial design (e.g., number or characteristics of subjects, duration of study, study arms/doses)
  - Example: what is the power of the X trial given Y design?

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- **General Modeling Process**
- General Simulation Process

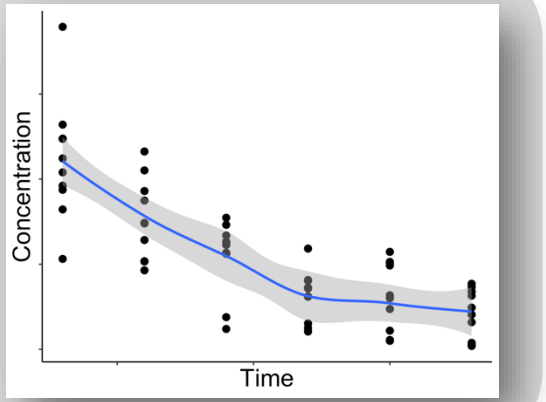
Model Development



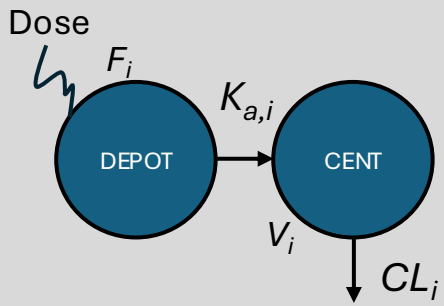
WT	AGE	SEX	...
65	31	F	...
77	25	M	...
102	36	M	...
...	...	...	...

ID	TIME	DV	AMT	CMT	EVID	WT	AGE
1	0	0	100	1	1	65	31
1	0.69	48.1	0	2	0	65	31
1	1.98	12.5	0	2	0	65	31
2	0	0	100	1	1	77	25
2	0.52	69.7	0	2	0	77	25
2	2.56	15.3	0	2	0	77	25

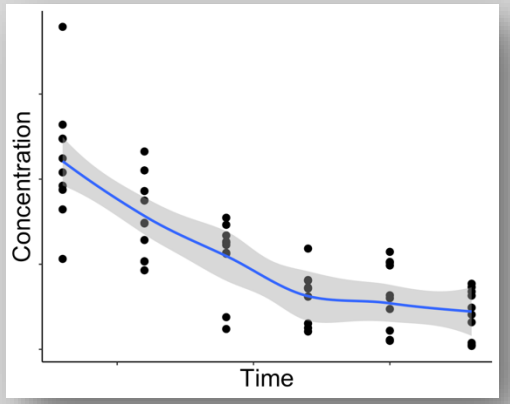
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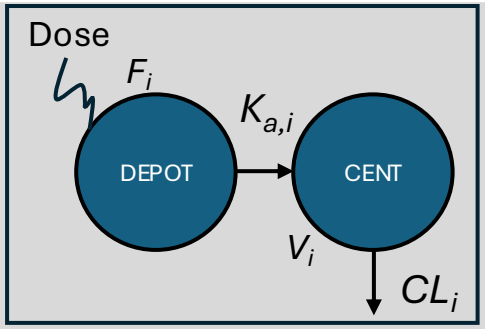
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Model Development



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$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\eta_1,i}$$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

...

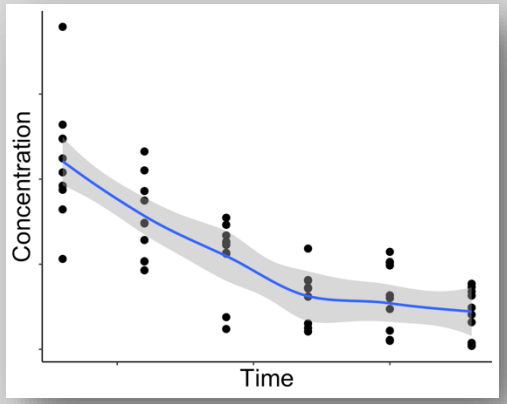
$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

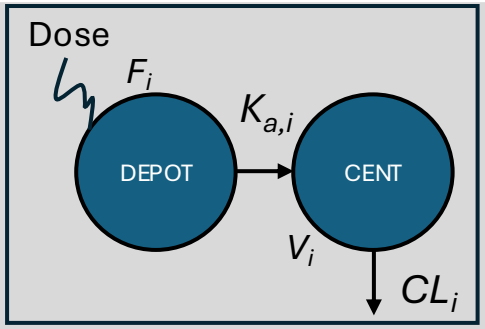
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Population parameters (`.ext`)

1. Fixed-effect

$\boldsymbol{\theta_1, \theta_2}$ ...

2. Random-effect

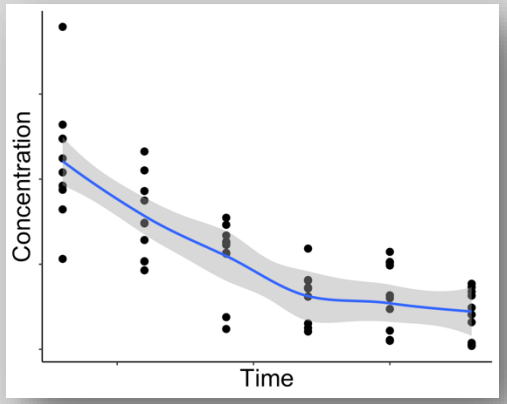
2.1 Ω matrix

$$\begin{pmatrix} \omega_{1,1}^2 & \cdots & \omega_{n,1}^2 \\ \vdots & \ddots & \vdots \\ \omega_{1,n}^2 & \cdots & \omega_{n,n}^2 \end{pmatrix}$$

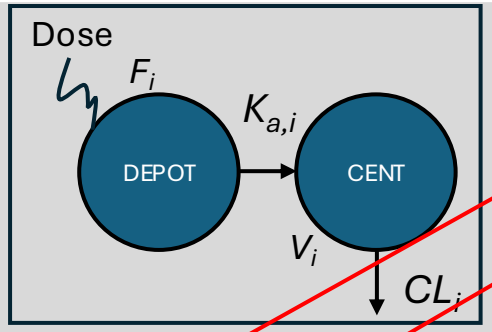
2.2 Σ matrix

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Model Development



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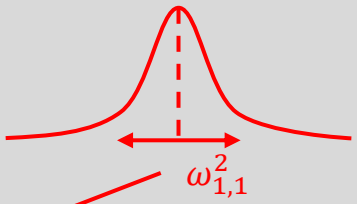
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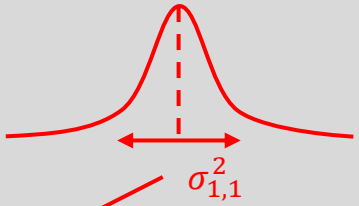
$\boldsymbol{\theta_1, \theta_2}$ ...

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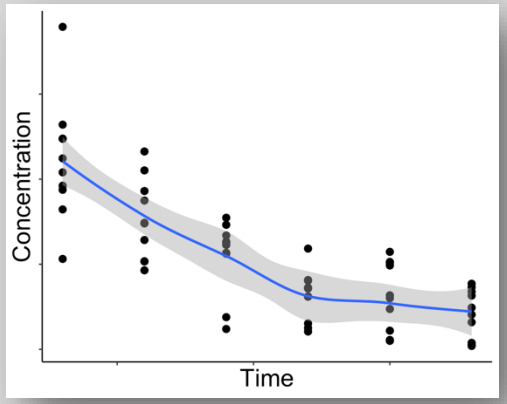
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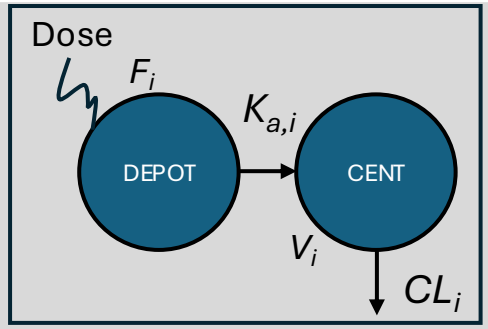
2.2 Σ matrix



Model Development



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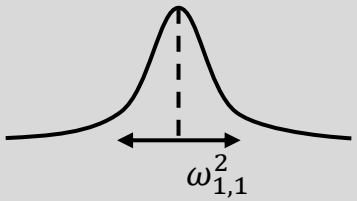
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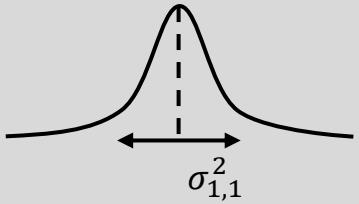
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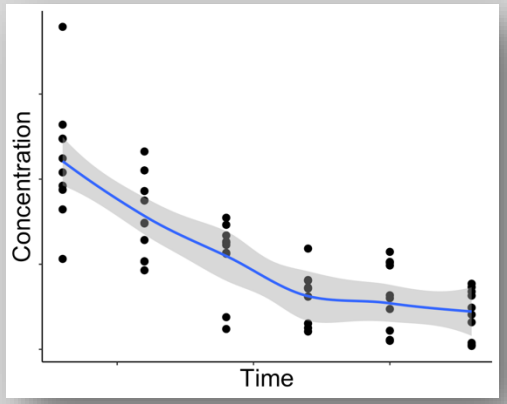


Individual random effects (`.phi`)
(Empirical-bayes estimates, EBE)

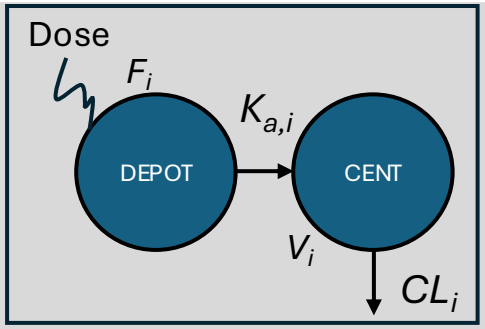
For each individual *i*:

$\boldsymbol{\eta_{1,i}, \eta_{2,i}}$….

Model Development



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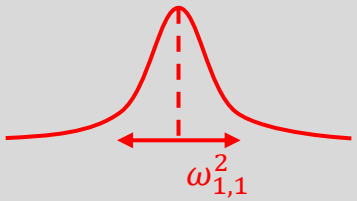
Population parameters (`.ext`)

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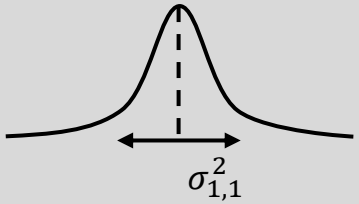
$\boldsymbol{\theta_1, \theta_2}$ …

2. Random-effect

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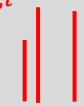
2.2 Σ matrix



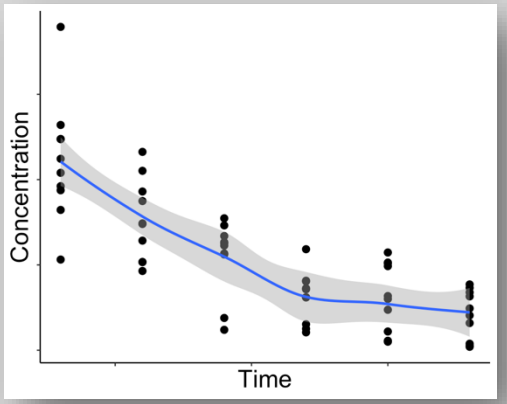
Individual random effects (`.phi`)

(Empirical-bayes estimates, EBE)

$\eta_{1,i}$



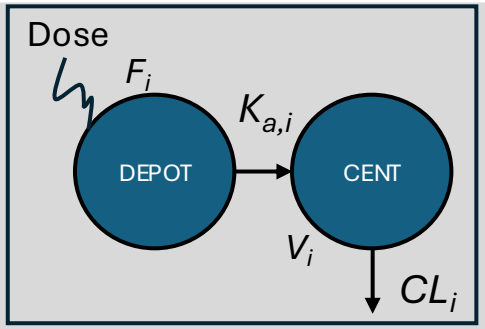
Model Development



WT	AGE	SEX	…
65	31	F	…
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…	…	…	…

Parameter precision (`.cov`)
(Variance-Covariance Matrix)

$$\begin{pmatrix} VAR(\theta_1) & \cdots & COV(\sigma^2_{n,n}, \theta_1) \\ \vdots & \ddots & \vdots \\ COV(\theta_1, \sigma^2_{n,n}) & \cdots & VAR(\sigma^2_{n,n}) \end{pmatrix}$$



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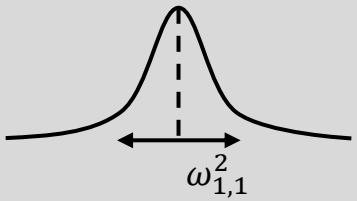
Population parameters (`.ext`)

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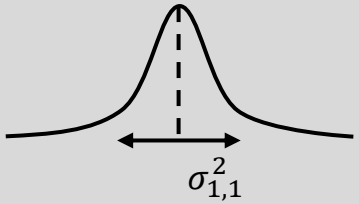
$\boldsymbol{\theta_1, \theta_2}$ …

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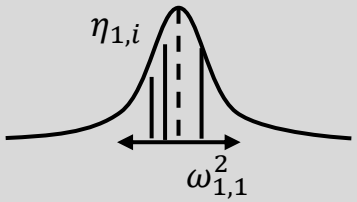
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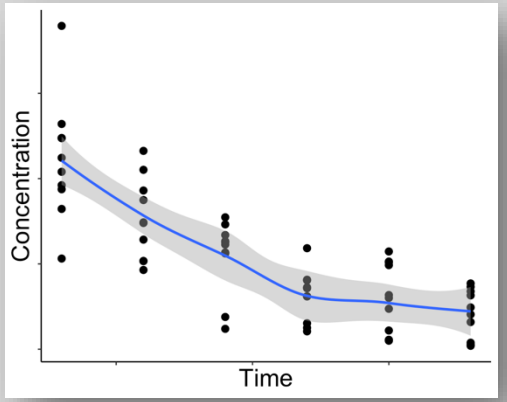
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Individual random effects (`.phi`)
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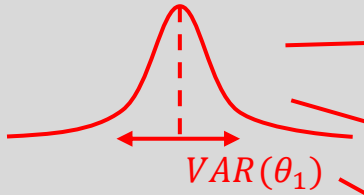


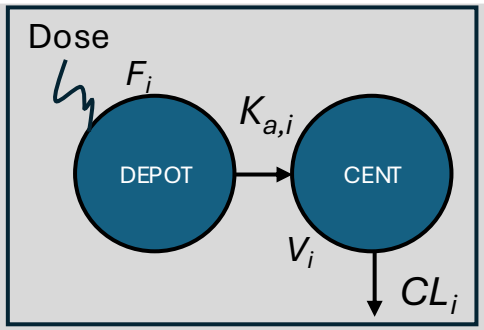
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Parameter precision (.cov)
(Variance-Covariance Matrix)





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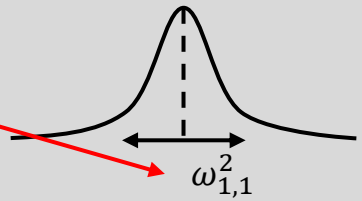
Population parameters (.ext)

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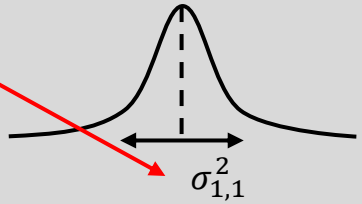
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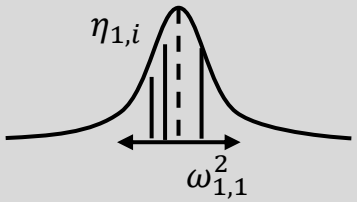
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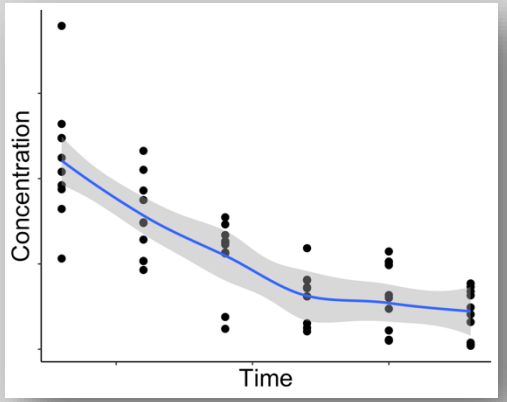
Individual random effects (.phi)
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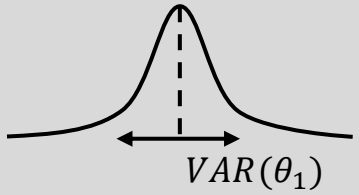
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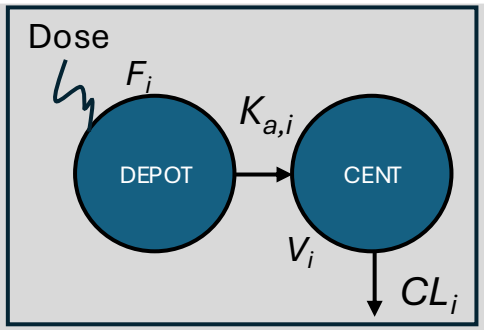
Simulation



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$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

…

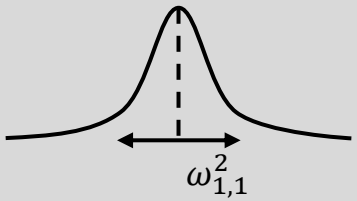
Population parameters (`.ext`)

1. Fixed-effect

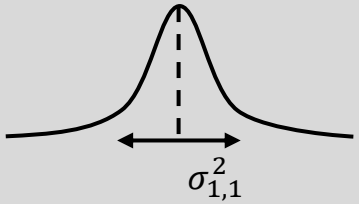
$\boldsymbol{\theta_1, \theta_2}$ …

2. Random-effect

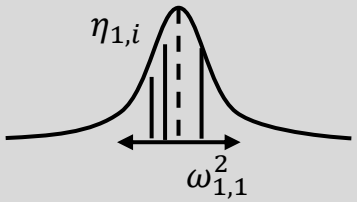
2.1 Ω matrix



2.2 Σ matrix



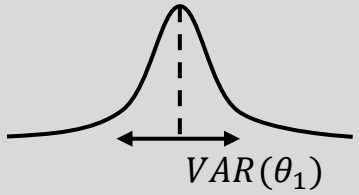
Individual random effects (`.phi`)
(Empirical-bayes estimates, EBE)

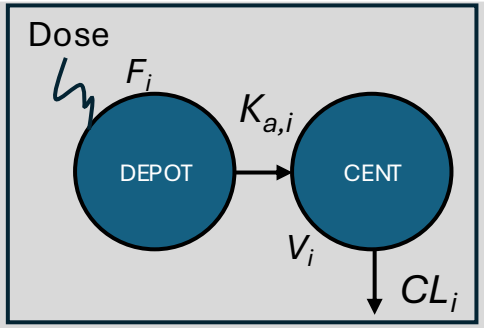


Simulation

WT	AGE	SEX	…
65	31	F	…
77	25	M	…
102	36	M	…
…	…	…	…

Parameter precision (`.cov`)
(Variance-Covariance Matrix)





$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\boldsymbol{\eta_{1,i}}}$$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

…

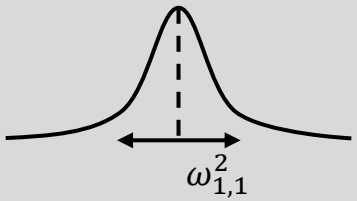
Population parameters (`.ext`)

1. Fixed-effect

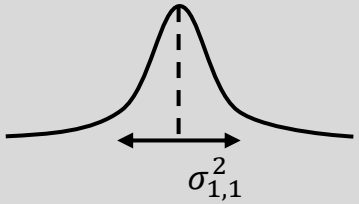
$\boldsymbol{\theta_1, \theta_2}$ …

2. Random-effect

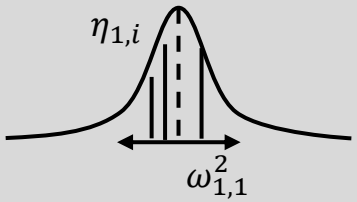
2.1 Ω matrix



2.2 Σ matrix

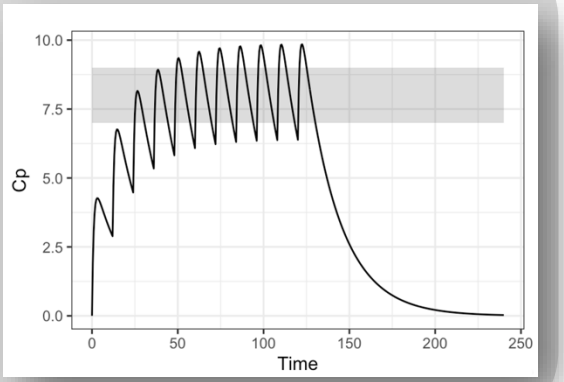


Individual random effects (`.phi`)
(Empirical-bayes estimates, EBE)



Simulation

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



WT	AGE	SEX	…
65	31	F	…
77	25	M	…
102	36	M	…
…	…	…	…

?

Dose

F_i

$K_{a,i}$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

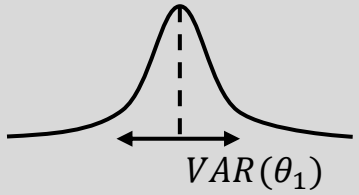
$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

…

The incorporation of simulation components depend on the type of simulation and the question to be answered

?

Parameter precision (`.cov`)
(Variance-Covariance Matrix)



?

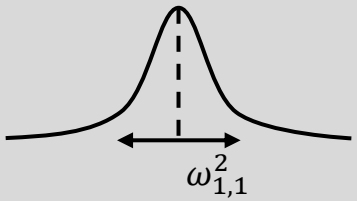
Population parameters (`.ext`)

1. Fixed-effect

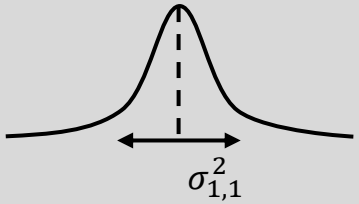
$\boldsymbol{\theta_1, \theta_2}$ …

2. Random-effect

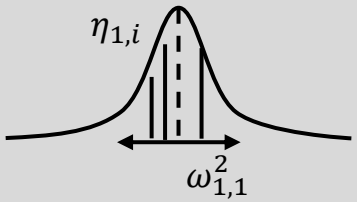
2.1 Ω matrix



2.2 Σ matrix

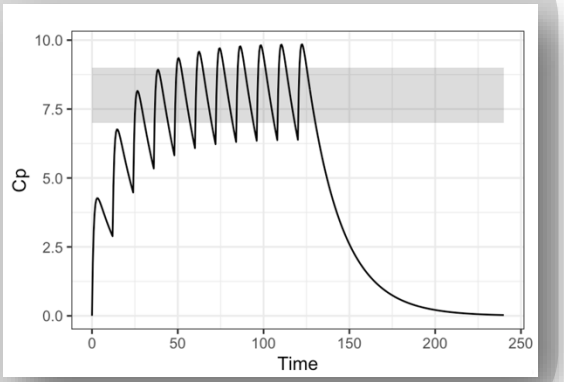


Individual random effects (`.phi`)
(Empirical-bayes estimates, EBE)

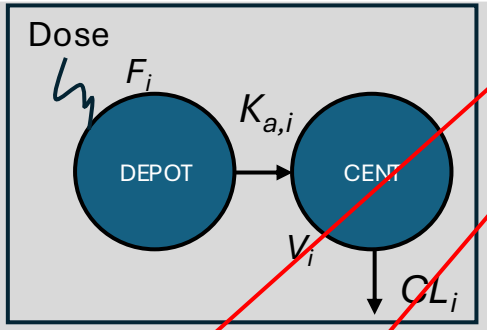


Deterministic Simulation-Typical Value Simulation

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



WT	AGE	SEX	…
65	31	F	…
77	25	M	…
102	36	M	…
…	…	…	…



$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\eta_{1,i}}$$

$$\eta_{1,i} = 0$$

…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

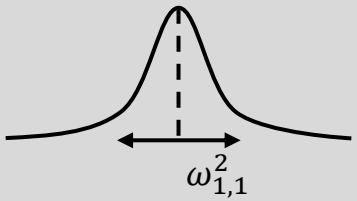
$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} = 0$$

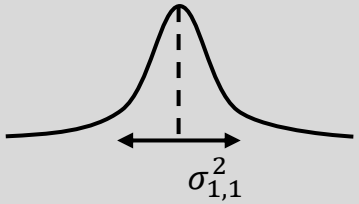
…

Population parameters (`.ext`)

1. Fixed-effect
   $\boldsymbol{\theta_1, \theta_2}$ …
2. Random-effect
   - 2.1 Ω matrix



   - 2.2 Σ matrix



Typical Workflow When Perform Typical Value Simulations

For i=1, …, nsub

Fix $\eta_i=0$ for subject i

Fix $\varepsilon_{ij}=0$ for subject i at time j

Derive $\theta_i = f(\theta, cov_i)$ for subject i

Calculate response $Y_{ij} = f(\theta_i, \eta_i=0, \varepsilon_{ij}=0)$

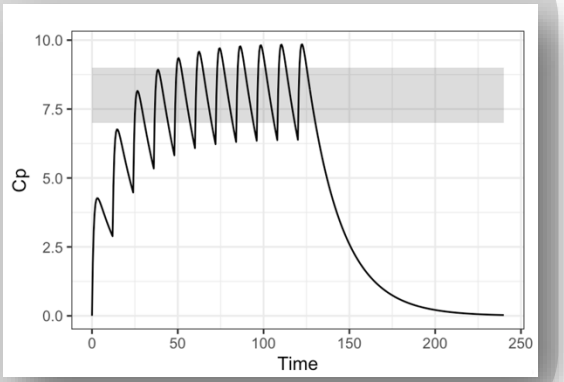
End

Adapted from Hu C. *J Pharmacokinet Pharmacodyn.* 2022 PMID: 35927373.

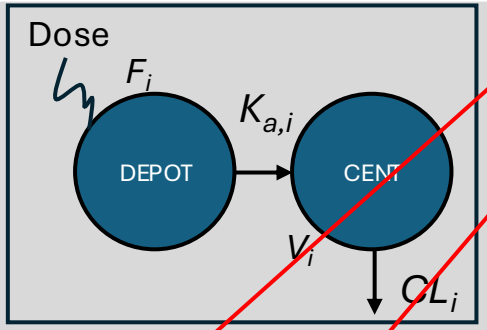
nsub: number of subjects
cov_i: covariates (e.g., body weight, age, etc) in subject i

Deterministic Simulation-EBE simulation

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



WT	AGE	SEX	...
65	31	F	...
77	25	M	...
102	36	M	...
...	...	...	...



$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\eta_{1,i}}$$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

...

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

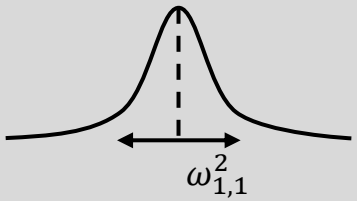
$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

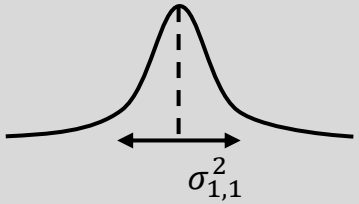
...

Population parameters (`.ext`)

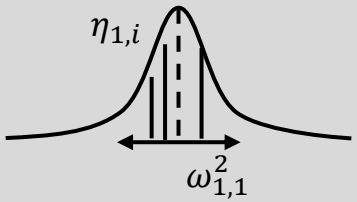
1. Fixed-effect
   $\boldsymbol{\theta_1, \theta_2}$ ...
2. Random-effect
   - 2.1 Ω matrix



   - 2.2 Σ matrix



Individual random effects (`.phi`)
(Empirical-bayes estimates, EBE)



Typical Workflow When Perform EBE Simulations

For i=1, …, nsub

Fix $\eta_i=\eta_i$ **in model output** for subject i

Fix $\varepsilon_{ij}=0$ for subject i at time j

Derive EBE_i= f(θ, cov_i, η_i) for subject i

Calculate response Y_{ij} = f(EBE_i,$\varepsilon_{ij}=0$)

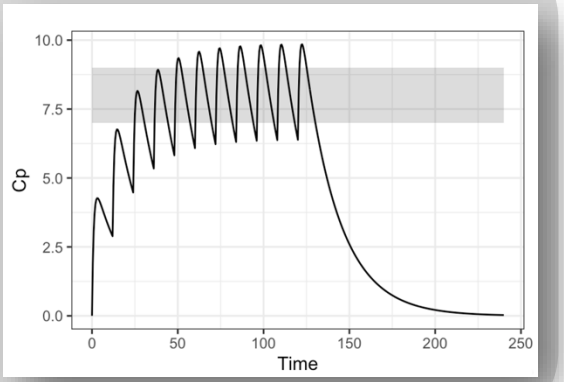
End

Adapted from Hu C. *J Pharmacokinet Pharmacodyn.* 2022 PMID: 35927373.

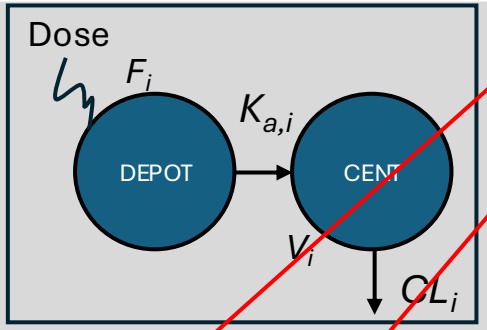
nsub: number of subjects
cov_i: covariates (e.g., body weight, age, etc) in subject i
EBE_i: empirical Bayes estimates in subject i

Stochastic Simulation

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



WT	AGE	SEX	…
65	31	F	…
77	25	M	…
102	36	M	…
…	…	…	…



$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\eta_{1,i}}$$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

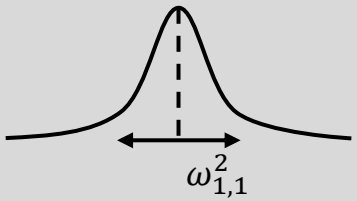
$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

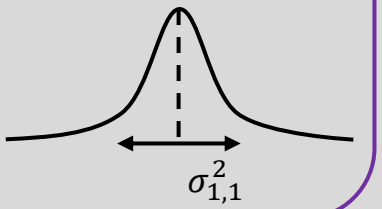
…

Population parameters (`.ext`)

1. Fixed-effect
   $\boldsymbol{\theta_1, \theta_2}$ …
2. Random-effect
   - 2.1 Ω matrix



   - 2.2 Σ matrix



Typical Workflow When Perform Stochastic Simulations

For i=1, …, nsub

Draw sample η_i for subject i using Ω matrix

*Draw sample ε_{ij} for subject i at time j using Σ matrix/Fix $\varepsilon_{ij}=0$

Derive $\theta_i = f(\theta, cov_i)$

Calculate response $Y_{ij} = f(\theta_i, \eta_i, \varepsilon_{ij})$

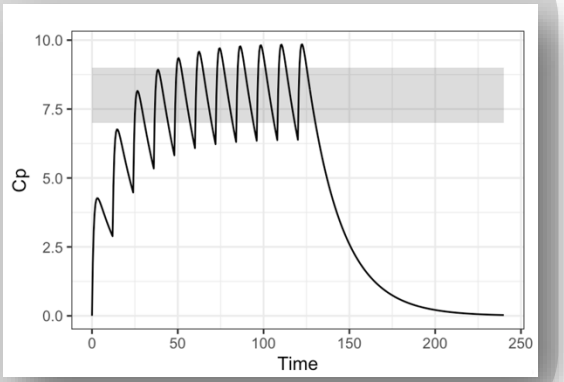
End

nsub: number of subjects
cov_i: covariates (e.g., body weight, age, etc) in subject i
*: this step may or may not be needed

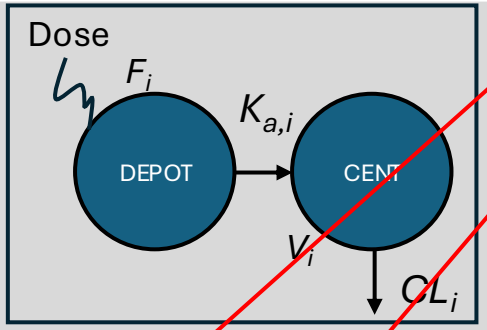
Adapted from Hu C. *J Pharmacokinet Pharmacodyn.* 2022 PMID: 35927373.

Covariate (Virtual Patient) Simulations

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



WT	AGE	SEX	…
45	12	M	…
57	15	F	…
33	6	F	…
…	…	…	…



$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\eta_{1,i}}$$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

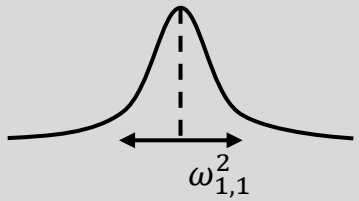
$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

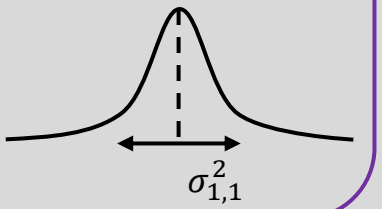
…

Population parameters (`.ext`)

1. Fixed-effect
   $\boldsymbol{\theta_1, \theta_2}$ …
2. Random-effect
   - 2.1 Ω matrix



   - 2.2 Σ matrix



Typical Workflow When Perform Stochastic Simulations

For $i=1, \ldots, nsub$

Draw sample η_i for subject i using Ω matrix

*Draw sample ε_{ij} for subject i at time j using Σ matrix/Fix $\varepsilon_{ij}=0$

Derive $\theta_i = f(\theta, cov_i)$

Calculate response $Y_{ij} = f(\theta_i, \eta_i, \varepsilon_{ij})$

End

nsub: number of subjects
cov_i: covariates (e.g., body weight, age, etc) in subject i
*: this step may or may not be needed

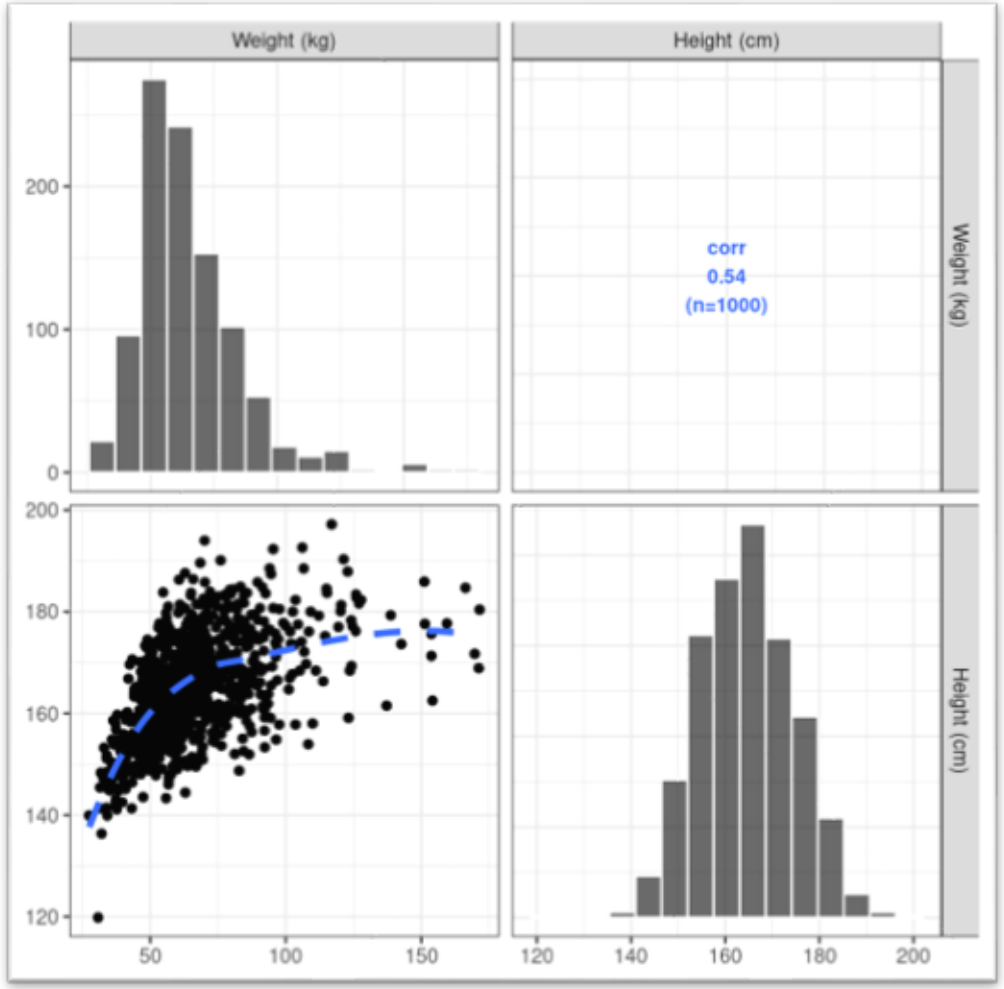
Adapted from Hu C. *J Pharmacokinet Pharmacodyn.* 2022 PMID: 35927373.

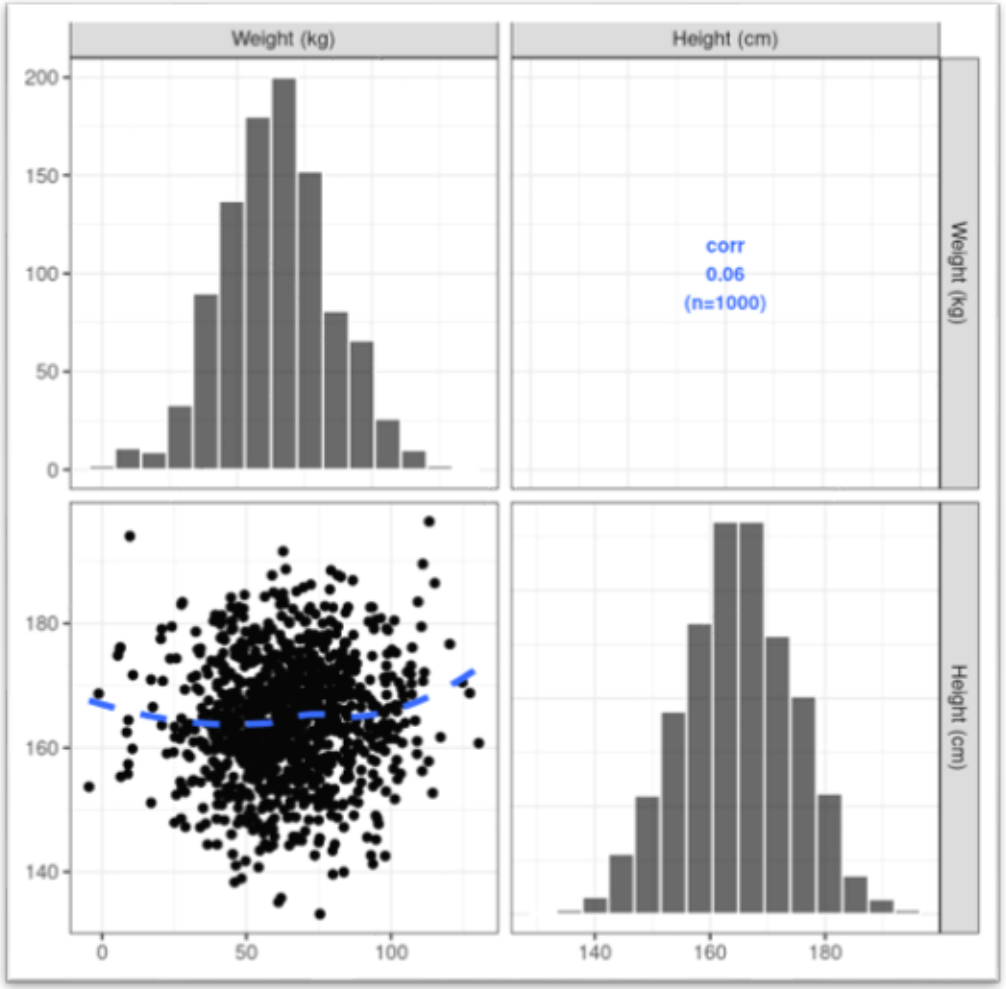
The Importance of Realistic Covariates/Virtual Patients

```
mean_WT  <- mean(test1$WT)
sd_WT    <- sd(test1$WT)
mean_HT <- mean(test1$HT)
sd_HT   <- sd(test1$HT)


withr::with_seed(1234,
                 test2 <- data.frame(ID=1:1000,
                                     WT=rnorm(1000,mean=mean_WT,sd=sd_WT),
                                     HT=rnorm(1000,mean=mean_HT,sd=sd_HT)
                                     )
                 )
```

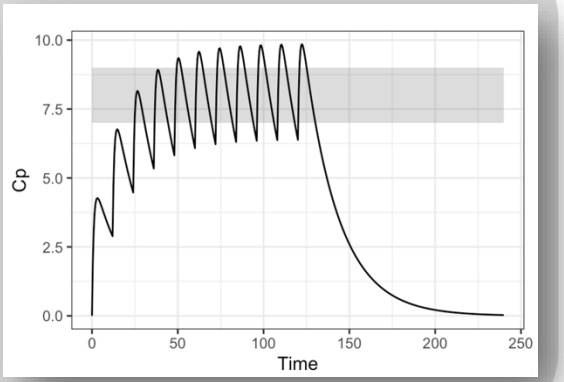
The Importance of Realistic Covariates/Virtual Patients



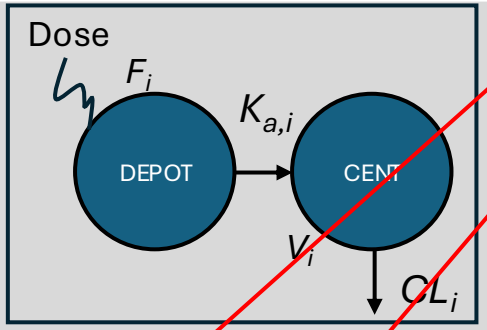


Stochastic Simulation-Multiple Replicates

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



WT	AGE	SEX	…
65	31	F	…
77	25	M	…
102	36	M	…
…	…	…	…



$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\eta_{1,i}}$$

$$\eta_{1,i} \sim N(0, \omega_{1,1}^2)$$

…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

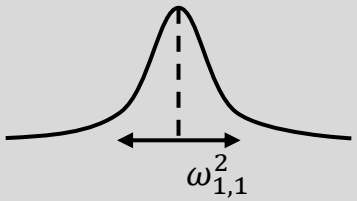
$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma_{1,1}^2)$$

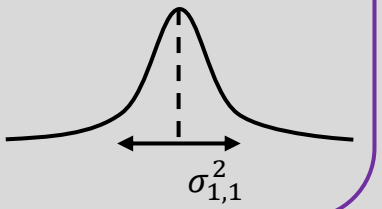
…

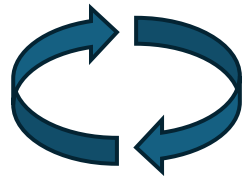
Population parameters (`.ext`)

1. Fixed-effect
   $\boldsymbol{\theta_1, \theta_2}$ …
2. Random-effect
   - 2.1 Ω matrix



   - 2.2 Σ matrix





n simulation replicates…

e.g., Visual Predictive Check (VPC)

Typical Workflow When Perform Stochastic Simulations with Replicates

For n=1, …, nrep

  Fixed population parameters (θ_n, Ω_n, Σ_n)

  For i=1, …, nsub

    Draw sample η_{ni} for subject i using Ω_n matrix

    *Draw sample ε_{nij} for subject i at time j using Σ_n matrix

    Derive $\theta_{ni} = f(\theta_n, cov_i)$

    Calculate response $Y_{nij} = f(\theta_{ni}, \eta_{ni}, \varepsilon_{nij})$

  End

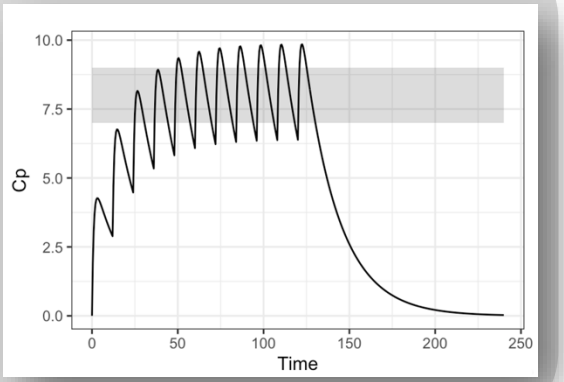
End

Adapted from Hu C. *J Pharmacokinet Pharmacodyn.* 2022 PMID: 35927373.

nrep: number of replicates
nsub: number of subjects
cov_i: covariates (e.g., body weight, age, etc) in subject i
*: this step may or may not be needed

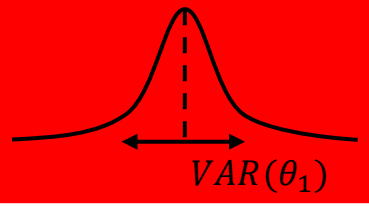
Simulations with Parameter Uncertainty

Simulation design:
- Treatment regimen
- Study duration
- Sampling schedule



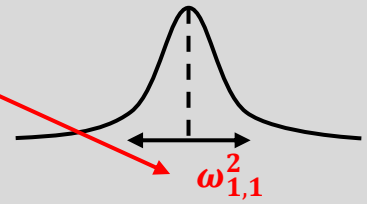
WT	AGE	SEX	…
65	31	F	…
77	25	M	…
102	36	M	…
…	…	…	…

Parameter precision (`.cov`)
(Variance-Covariance Matrix)

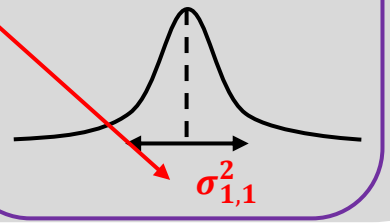


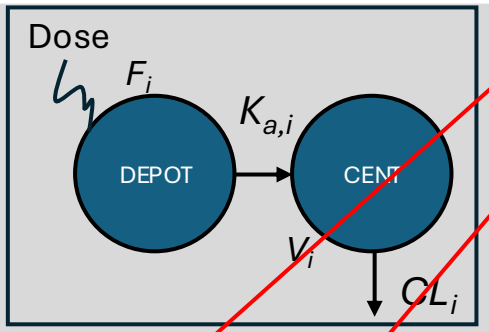
Population parameters (`.ext`)

1. Fixed-effect
   θ_1, θ_2 …
2. Random-effect
   - 2.1 Ω matrix



   - 2.2 Σ matrix





$$CL_i = \theta_1 \times \frac{WT_i}{70kg}^{\theta_2} \times e^{\eta_{1,i}}$$

$$\eta_{1,i} \sim N(0, \omega^2_{1,1})$$

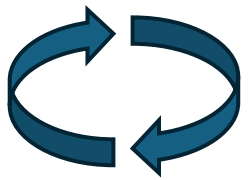
…

$$IPRED_{ij} = \frac{F_i \times Dose \times K_{a,i}}{V_i \times (K_{a,i} - \frac{CL_i}{V_i})} \times \left(e^{-\frac{CL_i}{V_i} \times t_{ij}} - e^{-K_{a,i} \times t_{ij}} \right)$$

$$OBS_{ij} = IPRED_{ij} + \epsilon_{1,ij}$$

$$\epsilon_{1,ij} \sim N(0, \sigma^2_{1,1})$$

…



n simulation replicates

n = number of (parameter vector) samples from uncertainty/precision distributions

Parameter uncertainty

- Parametric
  - Variance-Covariance matrix
    - Multivariate normal distribution (MVN) for fixed effects (θ)
    - Inverse wishart distribution for random effects (ω^2, σ^2)
- Non-parametric
  - Non-parametric bootstrap (BS)
  - Sampling-importance resampling (SIR)
  - Bayesian posterior distribution

Example-parameter uncertainty distribution

THETA26	THETA27	THETA28	THETA29	THETA30	THETA31	SIGMA11	SIGMA21	SIGMA22	SIGMA31	SIGMA32	SIGMA33	SIGMA41
78.2643	173.280	4.23748	11.29380	38.4408	0.421612	0.1284490	0	0.231501	0	0	1.42394e-02	0
75.4385	187.590	6.70469	10.08790	38.2204	0.770196	0.1230070	0	0.169305	0	0	5.43827e-02	0
79.6005	199.938	8.32571	8.66037	34.6556	0.459137	0.1171680	0	0.225894	0	0	2.33182e-02	0
87.9661	175.794	8.29748	11.13200	39.0711	0.759876	0.1244250	0	0.220386	0	0	2.17099e-02	0
75.8812	191.139	6.86712	9.22780	38.4680	0.732315	0.1244530	0	0.175997	0	0	9.39109e-02	0
70.3069	173.470	4.82680	14.95260	47.1471	0.579151	0.1076460	0	0.210610	0	0	7.75502e-02	0
81.8855	188.210	7.74528	11.30250	44.6379	0.600335	0.1237930	0	0.179454	0	0	3.97410e-02	0
66.3120	168.565	4.72414	10.55790	43.1807	0.689960	0.1142910	0	0.202239	0	0	1.01691e-01	0
81.8855	188.210	7.74528	11.30250	44.6379	0.600335	0.1237930	0	0.179454	0	0	3.97410e-02	0
79.9607	175.246	6.69603	13.90940	46.7332	0.495290	0.1135310	0	0.148579	0	0	9.13649e-02	0
74.4892	193.416	7.24374	10.24710	37.2643	0.544433	0.1077070	0	0.238335	0	0	3.82922e-02	0
74.3885	152.993	3.46359	16.16600	51.5856	0.767976	0.1066810	0	0.173881	0	0	1.10103e-01	0
73.3570	172.858	7.41016	15.44230	48.3020	0.593316	0.1251960	0	0.227475	0	0	4.25733e-02	0
78.1788	207.868	7.13567	7.37614	29.0706	0.386793	0.1120590	0	0.229197	0	0	1.10849e-02	0
70.425											e-02	0
72.453											e-02	0
70.1317	181.916	6.35709	10.07400	36.9336	0.614484	0.1305550	0	0.142311	0	0	3.75782e-02	0
75.3637	165.523	5.40291	13.32590	47.3354	0.711567	0.1088070	0	0.197540	0	0	8.80181e-02	0
81.8706	180.612	5.84744	14.58160	47.8523	0.500094	0.1133710	0	0.206229	0	0	1.66550e-02	0
80.1339	193.143	10.17130	8.30131	35.9364	0.524376	0.1077170	0	0.167458	0	0	4.54383e-02	0
72.2611	189.345	7.02739	10.94060	39.7045	0.590808	0.1170220	0	0.229699	0	0	3.47564e-02	0
87.4499	188.969	5.77477	9.64096	38.2691	0.400700	0.1050200	0	0.186759	0	0	7.05388e-02	0
76.3858	180.004	4.87934	9.24225	35.5641	0.553712	0.1105240	0	0.164901	0	0	7.51390e-02	0
76.9455	167.172	6.74738	8.96678	35.6594	0.703699	0.1093990	0	0.166276	0	0	6.86325e-02	0
66.5364	163.174	4.30772	12.83680	42.4348	1.139320	0.1069100	0	0.227534	0	0	1.81235e-02	0
89.0809	210.186	7.98011	7.84637	32.7716	0.532804	0.1216850	0	0.196216	0	0	2.99445e-02	0
75.4385	187.590	6.70469	10.08790	38.2204	0.770196	0.1230070	0	0.169305	0	0	5.43827e-02	0
79.6935	177.214	4.72871	11.53630	42.3676	0.517813	0.1200820	0	0.195516	0	0	2.70341e-02	0

1 row = 1 parameter vector =1 simulation replicate

Typical Workflow When Perform Simulations with Parameter Uncertainty

```
For n=1, …, nrep
    Draw a set of parameters (θn,Ωn,Σn) from the uncertainty distribution
    For i=1, …, nsub
        Draw sample ηni for subject i using Ωn matrix
        *Draw sample εnij for subject i at time j using Σn matrix
        Derive θni = f(θn,covi)
        Calculate response Ynij = f(θni,ηni,εnij)
    End
End
```

Adapted from Hu C. *J Pharmacokinet Pharmacodyn.* 2022 PMID: 35927373.

nrep: number of replicates
nsub: number of subjects
cov_i: covariates (e.g., body weight, age, etc) in subject i
*: this step may or may not be needed

The end